Interacting Multipoles of Rank 2 and 2:

$$T_{\alpha\beta\gamma\delta}\Theta_{\alpha\beta}\Theta_{\gamma\delta} = R_z^{-9} \cdot \frac{1}{9} \cdot \begin{bmatrix} +105 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Theta_{\gamma\delta} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot \delta\left(\gamma,\delta\right) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\gamma\delta} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot \delta\left(\beta,\delta\right) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\gamma\delta} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot \delta\left(\beta,\gamma\right) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\gamma\delta} \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot \delta\left(\alpha,\delta\right) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\gamma\delta} \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot \delta\left(\alpha,\gamma\right) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\gamma\delta} \\ -15 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot \delta\left(\alpha,\beta\right) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\gamma\delta} \\ +3 \cdot R_z^4 \cdot \delta\left(\alpha,\beta\right) \cdot \delta\left(\gamma,\delta\right) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\gamma\delta} \\ +3 \cdot R_z^4 \cdot \delta\left(\alpha,\gamma\right) \cdot \delta\left(\beta,\delta\right) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\gamma\delta} \\ +3 \cdot R_z^4 \cdot \delta\left(\alpha,\delta\right) \cdot \delta\left(\beta,\gamma\right) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\gamma\delta} \end{bmatrix} \tag{1}$$

Simplify deltas:

$$T_{\alpha\beta\gamma\delta}\Theta_{\alpha\beta}\Theta_{\gamma\delta} = R_z^{-9} \cdot \frac{1}{9} \cdot \begin{bmatrix} +105 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Theta_{\gamma\delta} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot \delta\left(\gamma,\gamma\right) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\gamma\gamma} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot \delta\left(\beta,\beta\right) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\beta\gamma} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot \delta\left(\beta,\beta\right) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\beta\delta} \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot \delta\left(\alpha,\alpha\right) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\alpha\gamma} \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot \delta\left(\alpha,\alpha\right) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\alpha\delta} \\ -15 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot \delta\left(\alpha,\alpha\right) \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\gamma\delta} \\ +3 \cdot R_z^4 \cdot \delta\left(\alpha,\alpha\right) \cdot \delta\left(\gamma,\gamma\right) \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\gamma\gamma} \\ +3 \cdot R_z^4 \cdot \delta\left(\alpha,\alpha\right) \cdot \delta\left(\beta,\beta\right) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\alpha\beta} \\ +3 \cdot R_z^4 \cdot \delta\left(\alpha,\alpha\right) \cdot \delta\left(\beta,\beta\right) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\alpha\beta} \end{bmatrix} \tag{2a}$$

$$\xRightarrow{\text{cleaned up}} R_z^{-9} \cdot \frac{1}{9} \cdot \begin{bmatrix} +105 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Theta_{\gamma\delta} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot \Theta_{\alpha\beta} \cdot \Theta_{\gamma\gamma} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot \Theta_{\alpha\beta} \cdot \Theta_{\beta\gamma} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Theta_{\beta\delta} \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot \Theta_{\alpha\beta} \cdot \Theta_{\alpha\gamma} \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Theta_{\alpha\delta} \\ -15 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\gamma\delta} \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\gamma\gamma} \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\beta} \cdot \Theta_{\alpha\beta} \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\beta} \cdot \Theta_{\alpha\beta} \end{bmatrix} \tag{2b}$$

Simplify R:

$$T_{\alpha\beta\gamma\delta}\Theta_{\alpha\beta}\Theta_{\gamma\delta} = R_z^{-9} \cdot \frac{1}{9} \cdot \begin{bmatrix} +105 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \Theta_{zz} \cdot \Theta_{\gamma\gamma} \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \Theta_{z\beta} \cdot \Theta_{z\beta} \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \Theta_{z\beta} \cdot \Theta_{z\beta} \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \Theta_{z\alpha} \cdot \Theta_{z\alpha} \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \Theta_{z\alpha} \cdot \Theta_{z\alpha} \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \Theta_{\alpha\alpha} \cdot \Theta_{zz} \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\gamma\gamma} \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\beta} \cdot \Theta_{\alpha\beta} \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\beta} \cdot \Theta_{\alpha\beta} \end{bmatrix} \tag{3a}$$

$$\xRightarrow{\text{cleaned up}} R_z^{-9} \cdot \frac{1}{9} \cdot \begin{bmatrix} +105 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{\gamma\gamma} \\ -15 \cdot R_z^4 \cdot \Theta_{z\beta} \cdot \Theta_{z\beta} \\ -15 \cdot R_z^4 \cdot \Theta_{z\beta} \cdot \Theta_{z\beta} \\ -15 \cdot R_z^4 \cdot \Theta_{z\alpha} \cdot \Theta_{z\alpha} \\ -15 \cdot R_z^4 \cdot \Theta_{z\alpha} \cdot \Theta_{z\alpha} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{\alpha\alpha} \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\gamma\gamma} \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\beta} \cdot \Theta_{\alpha\beta} \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\beta} \cdot \Theta_{\alpha\beta} \end{bmatrix} \tag{3b}$$

Simplify Multipoles:

$$T_{\alpha\beta\gamma\delta}\Theta_{\alpha\beta}\Theta_{\gamma\delta} = R_z^{-9} \cdot \frac{1}{9} \cdot \begin{bmatrix} +105 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{\gamma\gamma} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{\alpha\alpha} \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\gamma\gamma} \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\alpha\alpha} \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\alpha\alpha} \end{bmatrix} \tag{4a}$$

$$\xRightarrow{\text{cleaned up}} R_z^{-9} \cdot \frac{1}{9} \cdot \begin{bmatrix} +105 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{\gamma\gamma} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{\alpha\alpha} \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\gamma\gamma} \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\alpha\alpha} \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\alpha\alpha} \end{bmatrix} \tag{4b}$$

Exploit Traceless Conditions:

$$T_{\alpha\beta\gamma\delta}\Theta_{\alpha\beta}\Theta_{\gamma\delta} = R_z^{-9} \cdot \frac{1}{9} \cdot \begin{bmatrix} +105 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\alpha\alpha} \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\alpha\alpha} \end{bmatrix} \tag{5a}$$

$$\xRightarrow{\text{cleaned up}} R_z^{-9} \cdot \frac{1}{9} \cdot \begin{bmatrix} +105 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\alpha\alpha} \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\alpha\alpha} \end{bmatrix} \tag{5b}$$

Reduce Indices:

$$T_{\alpha\beta\gamma\delta}\Theta_{\alpha\beta}\Theta_{\gamma\delta} = R_z^{-9} \cdot \frac{1}{9} \cdot \begin{bmatrix} +105 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\alpha\alpha} \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\alpha\alpha} \end{bmatrix} \tag{6a}$$

$$\xRightarrow{\text{cleaned up}} R_z^{-9} \cdot \frac{1}{9} \cdot \begin{bmatrix} +105 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\alpha\alpha} \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\alpha\alpha} \end{bmatrix} \tag{6b}$$

Add up:

$$T_{\alpha\beta\gamma\delta}\Theta_{\alpha\beta}\Theta_{\gamma\delta} = R_z^{-9} \cdot \frac{1}{9} \cdot \begin{bmatrix} +45 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ +6 \cdot R_z^4 \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\alpha\alpha} \end{bmatrix} \tag{7}$$

Multiply Out Sum:

$$T_{\alpha\beta\gamma\delta}\Theta_{\alpha\beta}\Theta_{\gamma\delta} = R_z^{-9} \cdot \frac{1}{9} \cdot \begin{bmatrix} +45 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ +6 \cdot R_z^4 \cdot \Theta_{xx} \cdot \Theta_{xx} \\ +6 \cdot R_z^4 \cdot \Theta_{yy} \cdot \Theta_{yy} \\ +6 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \end{bmatrix} \tag{8a}$$

$$\xRightarrow{\text{added up}} R_z^{-9} \cdot \frac{1}{9} \cdot \begin{bmatrix} +51 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ +6 \cdot R_z^4 \cdot \Theta_{xx} \cdot \Theta_{xx} \\ +6 \cdot R_z^4 \cdot \Theta_{yy} \cdot \Theta_{yy} \end{bmatrix} \tag{8b}$$

Combining everything:

$$T_{\alpha\beta\gamma\delta}\Theta_{\alpha\beta}\Theta_{\gamma\delta} = \begin{bmatrix} +\frac{17}{3} \cdot R_z^{-5} \cdot \Theta_{zz} \cdot \Theta_{zz} \\ +\frac{2}{3} \cdot R_z^{-5} \cdot \Theta_{xx} \cdot \Theta_{xx} \\ +\frac{2}{3} \cdot R_z^{-5} \cdot \Theta_{yy} \cdot \Theta_{yy} \end{bmatrix} \tag{9}$$